Beyond Prevalence: Mitigation Signals and Interventions for Histopathology Patch Classification and WSI-Level MIL

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Abstract

A classifier can trail on a class for several distinct reasons, and mitigation methods differ in which of them they assume: some infer the need for mitigation from class frequency, others from example difficulty or from the within-class feature diversity a class exhibits. This report characterizes mitigation methods along two orthogonal axes: the signal that identifies a class or example as needing mitigation—prevalence, support, difficulty, or within-class diversity—and the level at which the intervention acts. Which methods apply depends on the input regime: labeled image patches support ordinary classification, whereas whole-slide image (WSI) diagnosis is commonly modeled as weakly supervised learning over patch-feature bags. Each method is therefore defined in the input form it consumes, with cross-entropy as the no-mitigation reference.

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1 Introduction

A pathology classifier can reach high aggregate accuracy while remaining unreliable on precisely the classes a diagnostic system is deployed to catch. When common tumor types dominate the training data, prevalence-sensitive evaluation rewards the classes that already dominate, so rare-class recall can degrade without the headline number moving. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware probability metrics as primary outcomes in imbalanced settings (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

Established mitigation methods differ not only in the level at which they intervene but also in the signal they use to infer which observations require mitigation. Class frequency is one such signal, and a low class count conflates several distinct shortages: a small share relative to the other classes, a small absolute amount of support, intrinsic difficulty under the available representation, and narrow diversity of the class's own feature manifold. Section 2 therefore makes both readings of the roster explicit, so that frequency is treated as a proxy for mitigation need rather than as its established cause. The methods themselves are drawn from the class-imbalance and long-tailed learning literature and are defined here in the order of their intervention level, stated in enough mathematical detail to make the assumed mechanism explicit. The aim is to fix the method definitions, assumptions, and fidelity qualifications used when instantiating these methods on specific pathology datasets.

Prediction regimes. Which methods apply depends on the input regime, so each method is defined in the input form it consumes. In the *patch* regime, each labeled crop is a training and prediction unit—not an independent observation, since crops from one case share its acquisition and much of its morphology (Section 2.3): a frozen foundation-model encoder such as Virchow2 embeds the patch once, and a lightweight classifier head is trained on the resulting embedding (Zimmermann et al., 2024; Vorontsov et al., 2024). In the *WSI-bag* regime, a slide is represented as a bag of frozen patch-feature instances that share one weak slide-level label, and an attention-based MIL head aggregates the instances into a slide prediction, through attention, instance ranking, or bag-level representation learning. Both regimes consume embeddings from the same pathology foundation-model family but differ in supervision unit, and freezing the encoder isolates imbalance interventions from end-to-end backbone learning. The broader computational-pathology pipeline these regimes sit in—whole-slide imaging, patching into tiles, and weakly supervised multiple-instance learning over frozen foundation-model features—is introduced in the companion datasets report (Queisler, 2026).

2 Taxonomy

Two questions organize the tested methods. The first asks which signal a method consults to infer *that* a class or an example needs help; the second asks at which level of the training pipeline it acts on that inference. The questions are independent, and neither answers the other: class-weighted CE gives example i a weight $w_i = f(n_{y_i})$ that depends only on how many training examples share its class, while focal loss gives it a weight $w_i = f(p_{y_i})$ that depends only on how confidently the model already predicts its label, so two methods that are both loss reweighting can rest on entirely different assumptions about what causes tail failure. This section fixes the notation used throughout and then defines both axes.

2.1 Notation

Let C be the number of target classes and let n_c denote the number of training examples from class c, so that $N = \sum_{j=1}^{C} n_j$ is the training-set size and $\pi_c = n_c/N$ is the empirical prevalence of class c. For a labeled patch or slide-level bag with label $y \in \{1, \ldots, C\}$, the model outputs logits $\mathbf{z} \in \mathbb{R}^C$ and class probabilities

$$p_c = \frac{\exp(z_c)}{\sum_{j=1}^{C} \exp(z_j)}. \tag{1}$$

For WSI-bag methods, a slide is represented as a bag $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ of frozen patch-feature instances. The attention-MIL backbone maps each bag to a bag embedding $\mathbf{b}(B)$ and class logits. Unless a method explicitly changes the sampler, objective, or representation loss, the empirical class frequencies in the training set define the optimization signal.

2.2 Intervention Levels

The intervention axis records where in the learning problem a method acts. Cross-entropy (CE) is the no-mitigation reference condition on this axis: it uses the observed training distribution as given and does not rebalance samples, labels, costs, or representations. The remaining methods group as data-level sampling, algorithm-level losses and prior corrections, hybrid sampling–loss methods, and representation or architecture-level methods (Saini and Susan, 2023; Zhang et al., 2023). Figure 1 places the tested roster in these families, and Section 3 defines the methods in that order, since a shared intervention level implies shared machinery to describe.

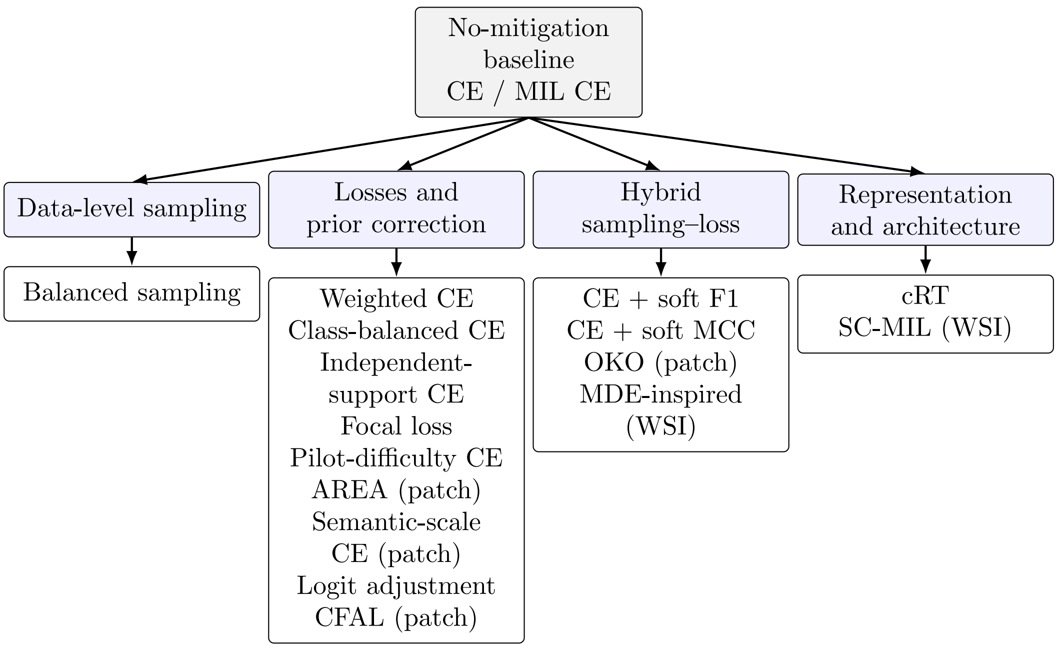

Figure 1: Intervention-level organization of the tested methods. Cross-entropy is the no-mitigation reference; alternatives intervene through sampling, objectives, prior correction, or representation and classifier design. Table 2 adds the orthogonal signal axis.

2.3 Mitigation Signals

The signal axis records what a method must observe before it concludes that mitigation is warranted, stated as the quantity the method actually consults rather than the property it ultimately affects.

Four signals are described in the following (Table 1): prevalence, support, difficulty, and diversity, with support splitting into a nominal and an independent reading. They are stated as properties of a class or example, independently of the intervention that acts on them.

Signal	What it describes	Representative quantity
Prevalence	Share of class c relative to the remaining classes	π_c, support rank, imbalance ratio ρ
Support	Absolute amount of material available for c, in either of two subtypes	*Nominal support*: n_c read absolutely rather than as a share, effective number $E_c(\beta)$. *Independent support*: distinct cases G_c or slides, refined by the redundancy-adjusted effective support $N_{\mathrm{eff},c}$
Difficulty	How hard a class or example is to classify under a fixed representation	Loss, predicted confidence p_y, decision margin, class difficulty d_c
Diversity	How much within-class variation the observed examples of c exhibit *under a specified representation*	Feature-manifold volume S'_c (semantic scale), effective area A_c, effective rank erank_c

Table 1: Signals a mitigation method can use to decide that a class or example needs help. Prevalence is relative and support is absolute, whether counted nominally or in independent units; difficulty and diversity are properties of the data rather than of the class-size profile, and both are read relative to a representation that has to be named with the estimate.

2.3.1 Prevalence

Prevalence is the share π_c of class c in the training distribution, together with the monotone summaries derived from it: the support rank of a class and the head-to-tail imbalance ratio $\rho = \max_c n_c / \min_c n_c$. It is a relative quantity. Rescaling every class count by a common factor leaves all prevalences unchanged, so a prevalence-aware rule behaves identically on a small and a large copy of the same class profile.

Prevalence is easily conflated with the amount of data a class actually carries, because both are usually read off the same counts n_c. The two come apart as soon as the profile changes shape rather than scale. Consider the training profiles $[8, 8, 8, 8]$ and $[64, 32, 16, 8]$. The scarcest class holds eight examples in both, so the material available for it is identical; its prevalence is $\frac{1}{4}$ in the first profile and $\frac{1}{15}$ in the second. A prevalence-aware rule treats that class very differently across the two profiles, whereas a rule driven by the absolute amount of material should treat it much the same. That absolute reading is the second signal.

2.3.2 Support

Support is the absolute amount of material a class carries, read as a magnitude rather than as a share. Two subtypes are kept apart throughout, because the roster contains methods of both kinds. *Nominal support* counts training examples: n_c read absolutely, or a saturating transform of it such as the effective number $E_c(\beta)$ of Equation (9), which still asks only how many examples there are. *Independent support* counts the units those examples came from, and the two readings coincide only when every example is its own unit.

In pathology they usually do not coincide, so independent support is not the raw example count. The material is nested (Figure 2): a cohort collects *cases*, a case is scanned into one or more slides, and a slide is tiled into patches. *Case* is used here as the patient-grouping identifier—the level holding all material originating from one patient, and the level splits are drawn over—rather than in the narrower clinical sense of a single specimen or accession. A patch is a training example, but patches drawn from one case are not independent observations of the class that case carries: they share its staining, scanner, preparation, and much of its morphology. Thousands of crops from a handful of cases therefore supply far less independent support than their nominal count suggests, and the same holds inside a slide bag, whose instances share a single weak label. Independent support is consequently counted in *split units*—the cases over which partitions are drawn—together with the distinct slides beneath them. At WSI level the labelled slide is the prediction unit, so a rare diagnosis represented by a few slides is poor in independent support however many instances those slides contain; whether it is also the independent unit depends on the cohort, since a case contributing several slides makes the case, not the slide, the unit clustering applies to.

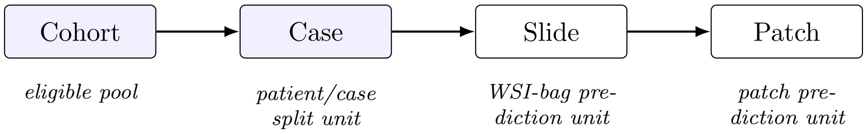

Figure 2: Nesting of the units independent support is counted in. Splits are drawn over cases, so slides and patches beneath one case are correlated rather than independent observations of its class.

Unique unit counts are the primary measure, but they are coarse: two classes with the same number of cases can still differ in how interchangeable their patches are. *Effective support* refines the count by discounting within-case redundancy. For class c, let G_c be the number of case or slide clusters contributing to it, let m_{gc} be their sizes, and let $n_c = \sum_g m_{gc}$. Redundancy is summarized by the intraclass correlation (ICC) of a scalar per-example score, computed from the between-cluster and within-cluster mean squares $\mathrm{MS}_{B,c}$ and $\mathrm{MS}_{W,c}$ with the usual correction $m_{0,c}$ for unequal cluster sizes:

$$m_{0,c} = \frac{n_c - \sum_{g=1}^{G_c} m_{gc}^2/n_c}{G_c - 1}, \qquad \mathrm{ICC}_c = \frac{\mathrm{MS}_{B,c} - \mathrm{MS}_{W,c}}{\mathrm{MS}_{B,c} + (m_{0,c} - 1)\mathrm{MS}_{W,c}}, \tag{2}$$

clipped to $[0, 1]$. An ICC near zero means that examples from one case are no more alike than examples from different cases; an ICC near one means the case is effectively a single observation repeated. The corresponding effective support follows the classical design-effect treatment of clustered observations, in which within-cluster correlation reduces the information carried by a nominal sample of n clustered observations by a design effect DE, leaving $N_{\text{eff}} = n/\mathrm{DE}$ (Kish, 1965; Rutterford et al., 2015). Under an exchangeable within-cluster correlation model with equally weighted observations, the cluster sizes m_{gc} are known here, so the design effect is used in its unequal-size form directly rather than through an equal-cluster approximation:

$$N_{\text{eff},c} = \frac{n_c}{\mathrm{DE}_c}, \qquad \mathrm{DE}_c = 1 + \big(m_c^\star - 1\big)\,\mathrm{ICC}_c, \qquad m_c^\star = \frac{\sum_{g=1}^{G_c} m_{gc}^2}{n_c}, \tag{3}$$

where the within-cluster correlation is estimated by ICC_c and $m_c^\star$ is the size-biased cluster size, the cluster size an average *observation* belongs to, which is what the design effect asks for. It relates to the plain mean cluster size $\overline{m}_c = n_c/G_c$ by $m_c^\star = \overline{m}_c\big(1 + \mathrm{CV}_c^2\big)$, with CV_c the coefficient of variation of the cluster sizes: the familiar form $1 + (\overline{m}_c - 1)\,\mathrm{ICC}_c$ is the equal-size special case, and the usual CV inflation of the design effect for spread-out cluster sizes is already contained in Equation (3) rather than being a separate correction. The two limits make the quantity readable: with $\mathrm{ICC}_c = 0$ the class retains its full count n_c, and with $\mathrm{ICC}_c = 1$ it collapses to $n_c/m_c^\star \leq G_c$, at most one observation per contributing cluster and fewer when the sizes are unequal. Equations (2)–(3) adjust a patch count for redundancy and are named *redundancy-adjusted effective support* throughout. The quantity is still read as an approximation rather than as an exact information content: distinct case and slide counts stay the primary independent-support measures, the exchangeable model treats every pair of examples inside a cluster as equally correlated regardless of where on the slide they were cut, and ICC_c depends on which per-example score it is estimated from. It is therefore reported with a sensitivity analysis over that choice of score, with classes whose $N_{\text{eff},c}$ ordering changes across scores flagged.

The two subtypes are what a method is read against. A method reads *nominal support* when n_c enters its rule absolutely rather than as the share π_c, and *independent support* when a unit count G_c or $N_{\mathrm{eff},c}$ does. Class-balanced CE models diminishing returns as a function of the raw count alone through $E_c(\beta)$ in Equation (9), and CFAL reuses the same construction; both are nominal-support methods, which separates them from the prevalence-driven methods, but neither asks how many independent units those examples came from. AREA is superficially close, since its effective area in Equation (18) deflates a count by the same design-effect algebra (Chen et al., 2023), but it measures redundancy among a class's features in the learned representation rather than over the case an example was cut from, which places it under diversity. Independent support is read by independent-support weighted CE (Section 3.3.3), a construction of this thesis that substitutes $N_{\mathrm{eff},c}$ for n_c inside the power-law weight of class-weighted CE.

2.3.3 Difficulty

Difficulty is how hard a class or an example is to classify under a fixed representation, independently of how often it occurs. At example level it is read from the model's own state during training: the current loss, the predicted probability p_y of the true class, or the margin between the true-class score and its strongest competitor. At class level it is measured as a probe error, for instance

$$d_c = 1 - \mathrm{recall}_c, \tag{4}$$

where recall_c is obtained from a class-balanced probe fitted on frozen features and evaluated out of fold. Balancing removes the relative class-frequency dominance that would otherwise let the probe objective be governed by the head classes, but it does not make the score independent of support: a class estimated from ten independent cases carries less information than one estimated from a hundred, whatever the sampling weights, so d_c mixes intrinsic separability with low-support estimation error. Equation (4) is therefore reported as *frequency-balanced* pilot difficulty. Isolating intrinsic difficulty under a fixed representation instead requires matched independent-unit support across classes, every class subsampled to the scarcest case count with the estimate averaged over repeated draws—the same matching condition Section 2.3.4 imposes on the diversity statistics, and for the same reason. A class whose morphology overlaps a neighbouring class is difficult even when its support is ample, which is what distinguishes this signal from the previous two, but only under matched support can the converse be asserted. Two readings are worth separating. Intrinsic difficulty is a property of the representation and holds across training conditions; condition-specific learnability describes what remains recoverable from one particular training set and therefore moves with allocated support. Example-level and class-level difficulty are also not interchangeable: a loss-based weight adapts continuously during training and can chase label noise, whereas d_c is estimated once and is stable but blind to which individual examples are hard.

2.3.4 Diversity

Diversity is how much within-class variation the observed examples of a class exhibit *under a specified representation*: the size of the region the class occupies in that feature space, read off those examples themselves. No count implies it. The qualification is the same one difficulty carries and is not a formality here—feature space is where the statistic is computed, so naming it is part of naming the quantity. Two classes with the same number of examples, contributed by the same number of independent cases, still differ in diversity when one presents a single growth pattern, grade, and preparation while the other presents several. Ma et al. (2023) make this the defining quantity of their *semantic scale*—the volume of a class's feature manifold—and show that classes of equal support can differ substantially in it, which is what motivates replacing the count by the measured volume as a reweighting signal (Section 3.3.7).

Diversity is summarized by statistics of the class's own features, all of them reference-free. Let $\lambda_1 \geq \cdots \geq \lambda_D$ be the eigenvalues of the covariance matrix of the class embeddings, each the variance of the class along one principal direction of the D-dimensional embedding space, and let $\widehat{\lambda}_i = \lambda_i / \sum_{j=1}^{D} \lambda_j$ be the same spectrum normalized to sum to one. With the Shannon entropy of that normalized spectrum,

$$H(\widehat{\lambda}) = -\sum_{i=1}^{D} \widehat{\lambda}_i \log \widehat{\lambda}_i, \qquad \mathrm{erank}_c = \exp\bigl(H(\widehat{\lambda})\bigr), \tag{5}$$

the effective rank erank_c counts how many directions the class occupies: it equals one when all variance lies along a single direction and D when the spectrum is flat (Roy and Vetterli, 2007). The participation ratio $\bigl(\sum_i \lambda_i\bigr)^2 / \sum_i \lambda_i^2$ summarizes the same spectrum with a different weighting. The two quantities that mitigation methods in this roster act on belong to the same family: the feature-manifold volume of semantic scale, Equation (20), is a log-determinant of that covariance and therefore weights the spectrum multiplicatively, and the effective area A_c of AREA, Equation (18), reads the same centred features through their mean pairwise correlation instead.

These statistics all inspect within-class feature geometry without being interchangeable summaries of it, so diversity is multidimensional. Effective rank reads the *normalized* spectrum and measures directional dimensionality alone: scaling every eigenvalue by a common factor leaves it unchanged, so a tight and a wide class occupying the same directions score alike. The semantic-scale volume responds to dimensionality and total spread together, and the effective area to mean pairwise redundancy, blind to how that redundancy is distributed over directions. Two classes can therefore share an effective rank and differ substantially in volume—which is what makes AREA against semantic-scale weighting a within-signal contrast between geometric summaries (Section 3.3.6), with erank_c carried alongside as a purely directional descriptor.

Two conditions keep such a statistic from re-measuring a signal already defined above. Diversity estimates are support-dependent in finite samples—a covariance estimated from few observations spans at most as many directions, and its spectrum is biased—and some estimators additionally contain the sample count explicitly, as A_c does through n_c in Equation (18). Comparisons therefore use matched support, by subsampling every class and averaging over draws; an unmatched estimate would let the class weight track n_c and reproduce prevalence or support under a different name. The matching is *hierarchical* rather than flat: every class contributes the same number of cases, and every selected case the same number of patches. A flat patch-level draw would let a case with ten thousand tiles dominate the covariance of its class, so its within-case morphology would be read as the diversity of the whole class; a case-level draw that collapsed each case to one summary would go too far in the other direction, since a biopsy can genuinely span several growth patterns, grades, and preparation artefacts. The hierarchical draw caps the influence of any one case while retaining the variation inside it. Note that this differs from how independent support is counted: there the case is the unit because correlated patches are not independent observations, whereas here patches from one case still carry feature-space variation and are kept, only balanced. The two signals describe different quantities and need not share an observational unit.

Which representation the statistic is read in is the third thing to fix, and it separates two readings of the signal exactly as intrinsic difficulty separates from condition-specific learnability. *Fixed diversity* is measured in the frozen Virchow2 embedding space: it is a property of the material under an encoder that no training condition alters, and erank_c is carried in that form as a locked predictor (Section 2.3.5). *Dynamic diversity* is measured in the 512-dimensional hidden representation of the head being trained, which is what the two acting methods read: the identical patches can therefore return different A_c and S'_c at epoch 1 and at epoch 20, because the representation moved and not the data. The dynamic reading is what makes the interventions adaptive, and it is also why their weights cannot be quoted as class constants—each is a class statistic indexed by a training step, and every reported value names the layer and the refresh it came from.

A fourth caution concerns what the signal cannot express. Diversity is computed from the observed material alone, so it reports how much space that material occupies and stays silent about variation the cohort never sampled: a class whose examples spread widely inside one morphological mode scores no worse than one that spreads equally widely across two. Diversity also depends on the training condition whenever support is itself manipulated, since subsampling a class changes which regions of it remain represented. Read as a property of the class, it must therefore be estimated on the eligible pool; read as a property of a training condition, it describes what that condition preserved. All these readings are reference-free, which is what allows a mitigation method to act on the signal during training, and two do: AREA (Section 3.3.6) estimates the space a class spans from the correlations among its own features and sets the class weight inversely to it, and semantic-scale weighted CE (Section 3.3.7) does the same from the volume of the class's feature manifold.

2.3.5 Reference Cohorts for Signal Estimation

Each signal above names a quantity; where it is estimated determines whether reading it is legitimate. Three cases arise. The scores locked before training—the frequency-balanced pilot difficulty d_c of Equation (4), the effective rank erank_c of Equation (5) when it is carried as a predictor, computed in the frozen Virchow2 embedding space as the fixed-diversity reading of Section 2.3.4, and the per-example scalar score that Equation (2) reads—are computed on a class-balanced reference manifest drawn from the development pool, with case-grouped out-of-fold cross-fitting so that no example contributes to the score it is scored by. The redundancy-adjusted effective support $N_{\mathrm{eff},c}$ is the one mixed case, and its two components are separated: the scalar score is locked as just described, and ICC_c, the cluster sizes m_{gc}, and hence $N_{\mathrm{eff},c}$ are then evaluated on the examples the training condition actually contains, using those locked scores. A condition therefore changes the effective support through its own cluster composition and within-cluster correlation, never through re-fitting the score the correlation is measured on. The dynamic scores A_c and S'_c are computed inside training from the current condition's own features and see nothing else. Selection quantities—hyperparameters, the tuned control of each method, and checkpoints—are read on the natural-validation split alone.

The test splits enter none of these. No statistic estimated on test data influences the training conditions, the class assignments, the hypotheses, the roster of included methods, or any hyperparameter; test logits are read once, at final evaluation, after every such decision is locked. This matters because a class-level statistic computed against a natural held-out cohort and then fed back into design decisions would be leakage regardless of how descriptive it looks, and per-class characterizations of the kind used here are exactly the quantities where that is easy to overlook. Any statistic that is reported on test data is therefore post-hoc characterization of a locked design and is labelled as such where it appears.

2.3.6 Signal and Intervention per Method

Signals are not to be confused with side effects. A method is treated as aware of a signal only when a quantity of that type enters its sampler, its objective, or its decision rule. Several methods reshape a property they never measure: SC-MIL compacts and separates bag embeddings, and the CFAL regularizer spreads class prototypes apart. Neither is triggered by an estimate of what the class fails to span, which separates them from AREA and semantic-scale weighted CE; both are triggered by class support or applied globally. Table 2 therefore records what each method consults alongside what it changes.

Counted by signal, eight interventions consult prevalence alone, and SC-MIL adds a partial case through class-aware batching. Two read difficulty: focal loss at example level and pilot-difficulty weighted CE at class level. Two read nominal support through the effective number, which CFAL combines with an example-level difficulty factor. Two read diversity, AREA and

Method	Mitigation signal	Intervention
Both regimes *patch and WSI-bag*		
CE, MIL CE	None; observed distribution used as given	None (reference)
Balanced sampling	Prevalence, through $n_{y_i}^{-s}$	Minibatch composition
Weighted CE	Prevalence, through n_c^{-s}	Per-example loss weight
Logit adjustment	Prevalence, through $\widehat{\pi}_c$ explicitly	Logit and prior correction
cRT	Prevalence, through balanced stage-two exposure	Classifier refit on a frozen representation
Class-balanced CE	Nominal support with diminishing returns, through the effective number $E_c(\beta)$	Per-example loss weight
Focal loss	Difficulty, through the per-example confidence p_y	Per-example loss weight
Pilot-difficulty weighted CE	Difficulty, through the locked frequency-balanced probe score d_c	Per-example class-dependent loss weight
Patch regime *only*		
Independent-support weighted CE	Independent support, through the redundancy-adjusted support $N_{\text{eff},c}$	Per-example class-dependent loss weight
AREA	Dynamic diversity, through the effective area A_c estimated from within-class feature correlations in the trained head representation	Per-example class-dependent loss weight
Semantic-scale weighted CE	Dynamic diversity, through the feature-manifold volume S_c' in the same representation; the inter-class crowding factor W_c is dropped, as in the source's DSB-CE-1 configuration	Per-example class-dependent loss weight
CFAL	Nominal support, through the effective number E_c of class c; difficulty, through $(1 - a_{i,y_i})^\gamma$; shapes diversity without measuring it	Prototype head and margin objective
CE + soft F1	Prevalence, through balanced sampling and equal classwise weight in the macro term	Sampler and auxiliary objective
CE + soft MCC	Prevalence, through balanced sampling; the batch marginals t_c, p_c of the MCC term are batch statistics, not prevalence measurements	Sampler and auxiliary objective
OKO	Prevalence, through uniform pair-class sampling	Set construction and set-aggregated loss
WSI-bag regime *only*		
SC-MIL	Label structure only; prevalence enters through class-aware batching; shapes diversity without measuring it	Contrastive term and CE curriculum
MDE-inspired	Prevalence, through paired natural and balanced loaders	Dual experts and consistency term

Table 2: Each tested method read along both axes: the signal it consults to identify mitigation need, and the mechanism it changes to act on it, grouped by the prediction regime the method applies to. Methods sharing an intervention family can consult different signals, and methods sharing a signal can intervene at different levels. Method-specific symbols name the mechanism responsible and are defined with the method in Section 3.

semantic-scale weighted CE, and one reads independent support, the construction of this thesis, since none of the collected methods counts the units a class's examples came from. Six interventions—class-weighted CE, class-balanced CE, independent-support weighted CE, pilot-difficulty weighted CE, AREA, and semantic-scale weighted CE—share a single intervention level: all six apply a mean-one normalized class weight to unmodified CE, and they differ in the per-class score they read, namely n_c, $E_c(\beta)$, $N_{\text{eff},c}$, d_c, A_c, and S'_c. Holding the intervention level fixed substantially reduces the mechanistic differences between them without eliminating them: the score enters through a different transform in each rule, and the two diversity rules additionally re-estimate their score during training after a deferred start, so they differ from the four fixed rules in schedule as well as signal. Contrasts among the prevalence-aware methods vary the intervention at a fixed signal, these six vary the signal at a largely fixed intervention, and difficulty and diversity each admit a within-signal contrast in addition. The quantities d_c, $N_{\text{eff},c}$, and erank_c are also carried forward as predictors in their own right, independently of whether a method acting on them succeeds.

3 Methods

Each method is defined below in the input form it consumes, following the intervention families of Figure 1. The comparison is regime-specific. Patch methods consume frozen Virchow2 embeddings of controlled patches and optimize one shared MLP classifier head; CE-based patch methods use the same linear output layer, while CFAL replaces that layer with class prototypes and affinity-based prediction. WSI-bag methods consume complete eligible frozen-feature bags and optimize one shared attention-MIL head. Both regimes therefore rely on the same pathology foundation-model family, but they differ in supervision unit: one embedding per selected patch versus aggregation over every eligible instance on a slide, without bag truncation.

3.1 Cross-Entropy Baseline

Unweighted cross-entropy is the baseline because it is the standard supervised objective without an explicit class-imbalance correction. For an example with label y, the loss is

$$\mathcal{L}_{\text{CE}}(\mathbf{z}, y) = -\log p_y. \tag{6}$$

In the patch regime, this objective trains the shared linear head on frozen Virchow2 patch embeddings from the controlled manifest. In the WSI-bag regime, the same objective trains the shared attention-MIL model on slide-level feature bags. In both cases, CE uses the observed class distribution directly: majority classes appear more often in minibatches and therefore contribute more loss terms over an epoch. This makes CE an appropriate reference for asking whether an imbalance-specific intervention actually improves the rare-class tradeoff.

3.2 Data-Level Sampling

Data-level methods change what the optimizer sees, rather than changing the mathematical penalty assigned to one observed example. In the taxonomy, this family includes the resampling strategies that make minority examples more visible during training (Saini and Susan, 2023).

3.2.1 Balanced Sampling

Balanced sampling changes minibatch composition by sampling training examples with probabilities related to their class support through a strength $s \in [0, 1]$. For an example i with class y_i, the sampler weight is proportional to

$$q_i(s) \propto n_{y_i}^{-s}. \tag{7}$$

After normalization across the training set, $s = 0$ recovers ordinary sampling and $s = 1$ gives equal expected exposure to every class; intermediate values partially rebalance the data. The patch variant applies this sampler to patch examples and still optimizes CE. The MIL variant applies the same idea at slide-bag level and also optimizes CE. Balanced sampling therefore tests a pure data-level intervention without changing the loss formula.

3.3 Algorithm-Level Losses

Algorithm-level methods keep the training examples fixed but alter how errors are penalized. They are cost-sensitive or difficulty-sensitive objectives: minority examples may receive larger explicit weights, or easy examples may contribute less to the gradient (Saini and Susan, 2023; Scholz et al., 2024).

3.3.1 Class-Weighted Cross-Entropy

Class-weighted CE multiplies each example's CE term by a class-dependent cost controlled by the same strength $s \in [0, 1]$:

$$\mathcal{L}_{\text{WCE}}(\mathbf{z}, y) = -w_y \log p_y, \qquad w_c(s) \propto n_c^{-s}. \tag{8}$$

The implementation normalizes these weights so that their mean across classes is one. Here $s = 0$ recovers unweighted CE and $s = 1$ gives inverse-frequency weighting; intermediate values reduce the correction. This method leaves the sample order and model architecture unchanged, but increases the loss contribution of classes with fewer training examples. It is evaluated for both patch classification and attention-MIL.

3.3.2 Class-Balanced Cross-Entropy by Effective Number

Class-balanced (CB) cross-entropy replaces inverse-frequency weighting by a saturating notion of how much a class count is worth (Cui et al., 2019). Its premise is that additional examples of a class need not contribute linearly increasing information: as support grows, examples increasingly resemble ones already seen, so the marginal contribution of the next example diminishes. For class c with training count n_c, the effective number of samples is

$$E_c(\beta) = \frac{1 - \beta^{n_c}}{1 - \beta}, \qquad 0 \leq \beta < 1, \tag{9}$$

and the class weight is proportional to its inverse. As for class-weighted CE, the implementation normalizes the weights to mean one across classes,

$$w_c^{\text{CB}}(\beta) = \frac{\widetilde{w}_c^{\text{CB}}(\beta)}{\frac{1}{C}\sum_{j=1}^{C} \widetilde{w}_j^{\text{CB}}(\beta)}, \qquad \widetilde{w}_c^{\text{CB}}(\beta) = \frac{1 - \beta}{1 - \beta^{n_c}}, \tag{10}$$

and optimizes

$$\mathcal{L}_{\text{CB}}(\mathbf{z}, y; \beta) = -w_y^{\text{CB}}(\beta) \log p_y. \tag{11}$$

The single control β sets the count scale at which those diminishing returns take effect. At $\beta = 0$ every effective number equals one and the objective reduces to ordinary CE; as $\beta \to 1$ the normalized weights approach inverse-count weighting. Intermediate values are the informative ones, because they separate a *saturating* count correction from the power-law correction n_c^{-s} of Equation (8): under the power law every doubling of class support changes the weight by the same factor, whereas under Equation (10) the weight stops responding once n_c exceeds roughly $(1 - \beta)^{-1}$. The reference default is $\beta = 0.999$; β is the imbalance control selected on validation.

Two properties make this method the cleanest available isolation of the effective-number mechanism. First, n_c enters absolutely rather than as a share, which makes CB-CE the nominal-support rule of the roster: unlike the prevalence-driven methods its weights change when a class profile is scaled rather than reshaped, and unlike independent-support weighted CE it asks only how many examples a class holds, not how many units they came from. Here n_c is the prediction-unit count of the training condition: labeled patches for patch classification and labeled slides for WSI-level MIL. Second, CB-CE changes neither the sampler, the head geometry, nor the output semantics, while CFAL couples the same effective number to class prototypes, a margin objective, an affinity-based difficulty factor, and a diversity regularizer (Equation (30)). An effect of CFAL therefore cannot be attributed to its effective-number term alone, whereas an effect of CB-CE can. The method is evaluated in both regimes.

3.3.3 Independent-Support Weighted Cross-Entropy

This method supplies the independent-support subtype of Section 2.3.2, against class-balanced CE as its nominal-support counterpart. Class-weighted CE and class-balanced CE both read the raw example count n_c, and AREA deflates it by feature-space redundancy rather than by the case a patch was cut from; the raw count in pathology overstates what a class carries, because patches drawn from one case share its staining, scanner, preparation, and much of its morphology (Section 2.3). Independent-support weighted CE is constructed for this thesis to supply that channel. The construction itself is not novel statistics: it transfers the design-effect correction for clustered sampling (Kish, 1965; Rutterford et al., 2015) into the class weight.

The rule keeps the functional form of Equation (8) and changes only the support measure it reads. With the redundancy-adjusted effective support $N_{\text{eff},c}$ of Equation (3) and the mean-one normalization used by the other weighting methods,

$$w_c^{\text{ISW}}(\tau) = \frac{N_{\text{eff},c}^{-\tau}}{\frac{1}{C}\sum_{j=1}^{C} N_{\text{eff},j}^{-\tau}}, \qquad \mathcal{L}_{\text{ISW}}(\mathbf{z}, y; \tau) = -w_y^{\text{ISW}}(\tau) \log p_y. \tag{12}$$

The exponent $\tau \geq 0$ is the single imbalance control, and $\tau = 0$ recovers ordinary CE. Because Equation (12) differs from class-weighted CE only in substituting $N_{\text{eff},c}$ for n_c, a difference between the two is attributable to the redundancy adjustment alone. Because $N_{\text{eff},c} = n_c/\text{DE}_c$ and both rules are normalized to mean one, they coincide at matched exponents exactly when $N_{\text{eff},c} = \kappa\, n_c$ for all c, that is when the design effect DE_c is constant across classes: the redundancy adjustment is then proportional and cancels in the normalization. They diverge when class-specific design effects differ, through cluster concentration, the spread of the cluster-size distribution, or the ICC. Equal case counts are consequently not sufficient for equivalence—two classes with the same number of cases can carry different cluster sizes or different within-cluster correlation and therefore different corrections. Both remain power-law corrections, which separates this method in turn from the saturating count correction of Section 3.3.2.

Two properties of the signal carry into the method. Effective support is a property of the training condition rather than of the class, so it is recomputed for each condition exactly as n_c is, in the split fixed by Section 2.3.5: the per-example scalar score is locked on the cross-fitted development reference, while ICC_c and the cluster sizes m_{gc} entering Equation (3) are evaluated on the condition's own examples. That scalar score is an experimental decision rather than part of the definition, so the weights are defined relative to it and inherit an estimation error class-weighted CE does not carry; the sensitivity analysis over that choice (Section 2.3) applies to them unchanged, and a class whose $N_{\text{eff},c}$ ordering is not stable under it cannot support a claim about the redundancy adjustment. The method is evaluated in the patch regime only. That restriction is conditional on the cohort rather than categorical: the collapse $m_c^\star = 1$, $N_{\text{eff},c} = n_c$ onto class-weighted CE holds when the labelled slide is itself the split unit, which requires that no case contribute several correlated slides. Where a case does contribute multiple slides, the clusters are cases and $m_c^\star > 1$, so WSI-level $N_{\text{eff},c}$ can still separate two classes with equal slide counts. The restriction to the patch regime is therefore a per-cohort decision: it holds for a slide-per-case cohort such as CAMELYON16, where slide-disjoint and case-disjoint splits coincide (Queisler, 2026), and would have to be reconsidered on a cohort whose cases contribute several slides.

3.3.4 Focal Loss

Focal loss reduces the contribution of confidently classified examples and concentrates optimization on examples that remain difficult. With focusing parameter γ, the multiclass focal objective is

$$\mathcal{L}_{\text{focal}}(\mathbf{z}, y) = -(1 - p_y)^\gamma \log p_y. \tag{13}$$

The reference default is $\gamma = 2.0$ for both regimes; γ is the tuned control selected on validation while the rest of the training protocol is held fixed. The method is relevant to imbalance because many minority-class examples can remain hard under a classifier dominated by head-class gradients, but the focal term does not directly encode the class counts. It is therefore a difficulty-sensitive loss rather than a sampler or explicit prior correction.

3.3.5 Pilot-Difficulty Weighted Cross-Entropy

Class-difficulty based weighting (CDB-W) replaces class frequency by measured classwise prediction difficulty as the signal for cost-sensitive learning (Sinha et al., 2022). It separates the assumption that a class is difficult because it is rare from the possibility that a comparatively frequent class remains intrinsically difficult under the available representation. In the source method, difficulty after epoch e is

$$d_{c,e} = 1 - A_{c,e}, \tag{14}$$

where $A_{c,e}$ is classwise accuracy on a separate balanced monitor set, and examples of class c receive the weight $d_{c,e}^{\tau_e}$, so that the objective at epoch e uses the difficulties measured at $e - 1$. The source additionally schedules the exponent τ_e from the imbalance of classwise performance, which makes CDB-W a dynamic class-difficulty method rather than a count-based reweighting rule. The mechanism differs from focal loss in the unit it describes: focal loss asks whether *this example* is currently hard through p_y, whereas CDB-W asks whether *this class* is currently hard. A frequent but morphologically ambiguous class can therefore be upweighted above a rare but already separable one.

Thesis adaptation. A source-faithful implementation requires a balanced monitor set that is re-evaluated throughout training. Sinha et al. (2022) note that this subset may double as the validation or development set, so reusing held-out data for it is consistent with CDB-W rather than excluded by it. The refusal to do so here is a benchmark-design choice: the natural-validation cohort is the only held-out data of that kind, and it is reserved for hyperparameter and checkpoint selection, so feeding it back as a per-epoch training signal would make model selection and training share a cohort across every method in the roster and break the comparison the benchmark is built on. The tested variant therefore uses the frequency-balanced class difficulty d_c of Equation (4), obtained once from the class-balanced out-of-fold probe and locked before the training conditions are constructed. With a fixed $\epsilon > 0$ that only prevents zero weights for a perfectly separated class, the mean-one normalized weight is

$$w_c^{\text{PDW}}(\tau) = \frac{(d_c + \epsilon)^\tau}{\frac{1}{C}\sum_{j=1}^{C}(d_j + \epsilon)^\tau}, \tag{15}$$

and the objective is

$$\mathcal{L}_{\text{PDW}}(\mathbf{z}, y; \tau) = -w_y^{\text{PDW}}(\tau) \log p_y. \tag{16}$$

The exponent $\tau \geq 0$ is the single imbalance control: $\tau = 0$ recovers ordinary CE, and larger values concentrate loss weight on the classes the balanced probe found hardest. Because this variant fixes difficulty in advance instead of tracking it during training, it is reported as *pilot-difficulty weighted CE (CDB-W-inspired)* rather than as a reproduction of CDB-W. It is not a dynamic method: it cannot react to a class whose difficulty changes as training proceeds, and it inherits whatever estimation error the probe carries, including the support-dependent component that balancing alone does not remove (Section 2.3).

That static reading is nevertheless the more informative one for a controlled comparison. Because d_c is estimated on a balanced reference before support is allocated, it stays fixed while the training conditions vary which classes are placed in the tail. Under a difficulty-aligned assignment the scarce classes are also the hard ones, so difficulty weighting and frequency weighting favor the same classes; under a difficulty-reversed assignment they issue opposite instructions. The contrast between this method and class-weighted CE therefore separates whether a class benefits from mitigation because it is rare or because it is hard, which a difficulty score re-estimated inside each condition would confound. The method leaves the sampler and architecture unchanged and is evaluated in both regimes.

3.3.6 AREA: Adaptive Reweighting via Effective Area

AREA weights classes by the size of the feature space they occupy rather than by how many examples they contain (Chen et al., 2023). Its premise is that a count is a poor description of a class's extent: a majority class is usually more diverse than its raw support suggests, so a weight derived from counts alone leaves the region a minority class should own compressed against its neighbours. The method estimates that extent non-parametrically from the correlations among a class's own samples, which makes it one of the two interventions in this roster whose class weight is set by a measured diversity statistic.

Let class c contribute the feature vectors $\mathbf{h}_1, \ldots, \mathbf{h}_{n_c}$ from the representation currently being trained, with class mean $\boldsymbol{\mu}_c$, and let $\mathbf{f}_i = \mathbf{h}_i - \boldsymbol{\mu}_c$ be those features centred on it. The pairwise correlation of the centred features is

$$R_{ij}^{(c)} = \frac{\langle \mathbf{f}_i, \mathbf{f}_j \rangle}{\|\mathbf{f}_i\| \, \|\mathbf{f}_j\|}, \tag{17}$$

and the *effective area* of the class is the inverse of the uniformly weighted quadratic form of that matrix, with $\mathbf{a} = \frac{1}{n_c}\mathbf{1}$:

$$A_c = \left(\mathbf{a}^\top R^{(c)} \mathbf{a}\right)^{-1} = \frac{n_c^2}{\sum_{i,j} R_{ij}^{(c)}} = \frac{n_c}{1 + (n_c - 1)\,\overline{r}_c}, \tag{18}$$

where $\overline{r}_c$ is the mean off-diagonal correlation. The right-hand form makes the estimator readable through $\overline{r}_c$ alone: a class whose centred features have zero *estimated* mean correlation has $A_c = n_c$ and keeps its full count, and A_c decreases monotonically as that estimate grows positive, reaching $A_c = 1$ only in the algebraic limit $\overline{r}_c = 1$. That limit is not the duplicate-example case, because centring on the sample mean forces $\sum_i \mathbf{f}_i = \mathbf{0}$: identical examples give $\mathbf{f}_i = \mathbf{0}$ for every i, which leaves Equation (17) undefined rather than equal to one, and nonzero centred vectors cannot all point in the same direction. The measured quantity is therefore how much positive redundancy remains among the centred features after the class mean is removed, not how close the raw examples are to copies of one another. The same constraint bounds the estimator from below: $\sum_{i,j}\langle \mathbf{f}_i, \mathbf{f}_j \rangle = \|\sum_i \mathbf{f}_i\|^2 = 0$, so $\overline{r}_c$ can be negative—with n_c centred vectors $\overline{r}_c \geq -1/(n_c - 1)$ is the only guarantee, and equal-norm centred features attain that bound exactly—in which case Equation (18) exceeds n_c and, at the bound, diverges. The tested implementation therefore clips $\overline{r}_c$ to $[0, 1]$ before forming A_c, so the effective area never exceeds the count and the weight stays finite; the clip is recorded because it makes the bound $A_c \leq n_c$

an implementation property rather than an algebraic one. Features with $\|\mathbf{f}_i\| = 0$ carry no direction and are excluded from Equation (17), with n_c in Equation (18) counting the retained ones; if fewer than two remain, the class spans no measurable extent at that refresh and A_c is set to its minimum of one. The class weight is inversely proportional to this area, normalized to mean one across classes as in the other weighting rules,

$$w_c^{\text{AREA}}(\tau) = \frac{A_c^{-\tau}}{\frac{1}{C}\sum_{j=1}^{C} A_j^{-\tau}}, \qquad \mathcal{L}_{\text{AREA}}(\mathbf{z}, y; \tau) = -w_y^{\text{AREA}}(\tau) \log p_y. \tag{19}$$

The exponent $\tau \geq 0$ is the single imbalance control: $\tau = 0$ recovers ordinary CE and $\tau = 1$ is the source rule. Because A_c is computed from the features the model currently produces, the weights follow the representation as it changes, which is what the method's name records.

Thesis adaptation. Three deviations from the source are recorded, so the method is reported as *AREA-inspired matched effective-area weighting* rather than as a reproduction of Chen et al. (2023). The weighting mechanism of Equations (17)–(19) is the source's, but all three deviations change the class statistic that mechanism is fed, not merely how it is computed: the estimator is pooled rather than accumulated, the sample it is estimated from is support-matched, and $\overline{r}_c$ is clipped to $[0, 1]$. The features entering Equation (17) are the 512-dimensional hidden representations of the shared MLP head, the same layer CFAL replaces with prototypes, and the weights are refreshed once per epoch after a deferred start at four fifths of the update budget U, matching the source's two-stage schedule of plain CE followed by area-weighted CE. The estimator itself is pooled rather than accumulated. The released implementation[1] computes an area per minibatch and sums those estimates over an epoch, so a class that appears in nearly every batch accumulates far more terms than a rare one and the resulting statistic partly tracks class occurrence rather than class extent. The tested variant instead estimates A_c once per refresh from the hierarchically matched subsample of Section 2.3.4: the same number of cases per class and the same number of patches per selected case. Matching is not optional in either version, since $A_c \leq n_c$ under the clipped estimator and an unmatched estimate would reproduce class size under a different name; balancing the draw across cases additionally keeps one heavily tiled case from setting a class's correlation structure, while the patches retained inside it still carry its genuine within-case variation. The clip is the third deviation, argued with Equation (18) above. Each is made to isolate diversity from support, which is the purpose the method serves in this roster; none is a neutral implementation detail, and results are therefore not read as evidence about the published method's behaviour under its own estimator.

Two contrasts follow from this placement. AREA carries one of the two class-level scores in the roster that move during training: n_c, $E_c(\beta)$, $N_{\text{eff},c}$, and d_c are all fixed before the first update, whereas A_c and the semantic-scale volume S'_c of Section 3.3.7 are re-estimated from the evolving representation, which reproduces at class level the static–dynamic distinction that separates pilot-difficulty weighted CE from focal loss. Against semantic-scale weighted CE, the difference is how the same class extent is summarized: A_c deflates the count by the mean pairwise correlation of the centred features, a first-moment summary of the Gram matrix that is blind to how that correlation is distributed over directions, whereas S'_c integrates the full covariance spectrum through a log-determinant. Both read within-class geometry only; the source's inter-class crowding factor is dropped precisely so that this contrast varies the statistic and nothing else. The method is evaluated in the patch regime only, since a rare diagnosis contributes too few bag embeddings for a stable correlation estimate at matched size.

[1] `https://github.com/xiaohua-chen/AREA`, accessed 14 August 2026.

3.3.7 Semantic-Scale Weighted Cross-Entropy

Semantic-scale-balanced learning weights classes by the volume of the feature manifold they occupy (Ma et al., 2023). Its premise is stated directly as a diversity claim: sample number fails to describe how much a class actually carries, because classes differ in feature richness, so the quantity a reweighting rule should read is the within-class diversity itself. That quantity is the class's *semantic scale*, and the method replaces the count by it everywhere the count would otherwise appear.

Let class c contribute the feature vectors $\mathbf{h}_1, \ldots, \mathbf{h}_{n_c}$ from the representation currently being trained, with class mean $\boldsymbol{\mu}_c$, and collect the centred features $\mathbf{f}_i = \mathbf{h}_i - \boldsymbol{\mu}_c$ as the columns of $Z_c \in \mathbb{R}^{d \times n_c}$. The volume of the region these features span is proportional to the determinant of their covariance, $\det\big(\frac{1}{n_c} Z_c Z_c^\top\big)$, which vanishes whenever the sample covariance is rank-deficient—the ordinary case here, since $n_c < d$ after the matched-size subsampling required of any diversity statistic. Following the rate-distortion construction the method borrows, the volume is therefore measured at a finite precision ε, which regularizes the determinant and yields the semantic scale

$$S'_c = \tfrac{1}{2} \log_2 \det\Big(I + \tfrac{d}{n_c \varepsilon^2} Z_c Z_c^\top\Big). \tag{20}$$

The precision is carried explicitly here. Ma et al. (2023) derive the same expression with ε in it, but their operational formula and pseudocode drop the factor and read $\frac{1}{2} \log_2 \det\big(I + \frac{d}{n_c} Z_c Z_c^\top\big)$, on the argument that the sphere radius does not change the relative size of the spaces the classes span; the surrounding text nonetheless states $\varepsilon = 1{,}000$. Equation (20) is therefore the derivation's form rather than the published operational one, with $\varepsilon = 1$ recovering the latter exactly, and the parameter is retained as a thesis-side regularization because the scale of the hidden representation here is not the source's. It is held fixed rather than tuned (see below). Equation (20) aggregates the same eigenvalues that Equation (5) does, through $\sum_i \log_2(1 + \frac{d}{n_c \varepsilon^2} \lambda_i)$ rather than through their entropy: a class occupying many directions with comparable variance scores high, and one whose variance collapses onto few directions scores low however many examples it contributes.

Raw volume is subject to a marginal effect. Each additional example is increasingly likely to fall in directions the class already spans, so S'_c saturates as n_c grows and understates how differently two classes are supported. The method therefore reads volume relative to the space a class is left by its neighbours. With class centres $\mathbf{o}_1, \ldots, \mathbf{o}_C$, let

$$w'_c = \frac{1}{C-1} \sum_{j \neq c} \|\mathbf{o}_c - \mathbf{o}_j\|_2, \qquad W_c = \log_2\big(\alpha + \widetilde{w}'_c\big), \qquad \alpha \geq 1, \tag{21}$$

where $\widetilde{w}'_c$ is w'_c normalized by its maximum across classes, so that an isolated class receives a larger factor than one crowded by similar classes. The source's relative semantic scale is $S_c = S'_c W_c$, with the smoothing parameter α setting how much of it the inter-class term is allowed to explain: large α flattens W_c and returns S_c to the pure volume S'_c.

The tested rule reads S'_c alone. The two factors of S_c measure different things: S'_c is a within-class quantity, the volume the class occupies by itself, whereas W_c is a function of distances between class *centres* and therefore describes how crowded a class is by its neighbours, which is a separability property and belongs to the difficulty signal rather than to diversity. A weight built on S_c would consequently be a mixed diversity-and-separability intervention and could not be contrasted against AREA at a fixed signal, which is the contrast this roster is built to support. Dropping W_c is not an unsupported truncation of the source: Ma et al. (2023) evaluate exactly this variant in their appendix, as the dynamic semantic-scale-balanced losses without inter-class interference denoted DSB-CE-1 and DSB-Focal-1, and report almost no performance degradation relative to the full DSB-CE and DSB-Focal. The pure within-class rule is therefore a published configuration of the method rather than a thesis invention, which is what makes it usable as the diversity arm of the signal comparison. The class weight is therefore inversely proportional to the volume alone, normalized to mean one across classes as in the other weighting rules,

$$w_c^{\mathrm{SSB}}(\tau) = \frac{S_c'^{-\tau}}{\frac{1}{C}\sum_{j=1}^{C} S_j'^{-\tau}}, \qquad \mathcal{L}_{\mathrm{SSB}}(\mathbf{z}, y; \tau) = -w_y^{\mathrm{SSB}}(\tau) \log p_y. \tag{22}$$

The exponent $\tau \geq 0$ is the single imbalance control: $\tau = 0$ recovers ordinary CE, and $\tau = 1$ matches the source's multiplier form $1/S_c$ with W_c suppressed, which is what the source itself approaches as α grows. Because S_c' is computed from the features the model currently produces, the weights follow the representation as it changes, which is what makes the source method a dynamic reweighting framework rather than a fixed cost matrix.

Thesis adaptation. The tested rule follows the source's W_c-free configuration, so it is reported as *semantic-scale weighted CE (DSB-CE-1-inspired)*; the qualifier records two further departures from Ma et al. (2023), the estimator and the retained precision parameter.

The estimator is changed exactly as for AREA. The features entering Equation (20) are the 512-dimensional hidden representations of the shared MLP head, and S_c' is refreshed once per epoch after a deferred start at four fifths of the update budget U, from the hierarchically matched subsample of Section 2.3.4: the same number of cases per class and the same number of patches per case. The source instead maintains a feature pool of recent minibatches, updated every iteration and reweighting only after a warm-up phase, which makes the estimate depend on how often a class appeared. That dependence is deliberate there: the marginal effect of Equation (21)'s motivation is part of the source's argument, and the source quantity intentionally combines observed support with feature diversity, so a class that is both scarce and narrow is penalized on both counts at once. The matched estimator used here orthogonalizes the latter from the former, because the signal-isolation experiment this roster is built for requires diversity to vary while support is held fixed—an unmatched pool would let S_c' grow with the number of features it is computed from and recover class size under a different name, collapsing the diversity arm onto the support arm. The adaptation therefore removes a component the source wanted, for a reason the source did not have. Fixing the refresh schedule and the subsampling to AREA's additionally leaves the statistic itself as the only difference between the two diversity rules, which is what makes the within-signal contrast of Section 2.3 interpretable.

The precision ε is the remaining departure: it is held fixed after standardizing the feature coordinates rather than being set to the source's implicit $\varepsilon = 1$ on unstandardized features, and it is not tuned, leaving τ as the imbalance control selected on validation. The standardization is a single global coordinate transformation, estimated once over all classes and applied identically to every class. Standardizing per class would rescale each class's own axes and thereby erase part of the total-spread difference S_c' exists to measure, leaving something closer to a purely directional statistic like erank_c. The parameter α has no role once W_c is dropped. The method is evaluated in the patch regime only, for the same reason as AREA: a rare diagnosis contributes too few bag embeddings for a stable covariance estimate at matched size. The paper's second use of semantic scale, as a label-free diagnostic of model bias, is not tested here.

3.3.8 Logit Adjustment

Logit adjustment explicitly corrects for the empirical training prior (Menon et al., 2021). Let

$$\widehat{\pi}_c = \frac{n_c}{\sum_{j=1}^{C} n_j} \tag{23}$$

denote the class prior estimated only from the current training condition. The control $\tau \geq 0$ determines the strength of the correction; $\tau = 0$ recovers ordinary CE. In the train-time form, adjusted logits are used inside the loss,

$$\mathcal{L}_{\mathrm{LA-train}}(\mathbf{z}, y) = -\log \frac{\exp(z_y + \tau \log \widehat{\pi}_y)}{\sum_{c=1}^{C} \exp(z_c + \tau \log \widehat{\pi}_c)}. \tag{24}$$

The positive prior term makes head classes harder to fit during training, inducing a classifier whose unadjusted logits favor a class-balanced target. In the post-hoc form, a model is trained with ordinary CE and corrected only at inference,

$$p_c^{\mathrm{LA}}(\mathbf{x}) = \frac{\exp(z_c(\mathbf{x}) - \tau \log \widehat{\pi}_c)}{\sum_{j=1}^{C} \exp(z_j(\mathbf{x}) - \tau \log \widehat{\pi}_j)}. \tag{25}$$

The sign difference follows from where the correction is applied: the train-time loss learns against prior-shifted logits, whereas the post-hoc rule removes the observed training prior from ordinary CE logits. Both patch and WSI regimes use the same definition at their respective prediction unit. Because either form deliberately changes the implied target prior, its probabilities should not be interpreted as calibrated for the natural test prevalence without an explicit target-prior assumption. Discrimination and calibration are therefore reported together, with post-hoc temperature scaling kept conceptually separate from prior correction.

Score variants with different interpretations are retained as separate outputs. For an ordinary CE score z_c^{CE}, post-hoc logit adjustment uses $z_c^{\mathrm{CE}} - \tau \log \widehat{\pi}_c$ for class-balanced decisions; this score is not reported as a probability calibrated to the natural target prevalence. Under the prior-shift assumption, probabilities for a target prior π_c^{target} instead normalize the logits

$$z_c^{\mathrm{CE}} - \log \widehat{\pi}_c + \log \pi_c^{\mathrm{target}}. \tag{26}$$

For train-time adjustment, CE is applied to $g_c = z_c + \tau \log \widehat{\pi}_c$. Because g_c approximates training-posterior logits, the corresponding target-prevalence logits are

$$z_c + (\tau - 1) \log \widehat{\pi}_c + \log \pi_c^{\mathrm{target}}. \tag{27}$$

The unadjusted z_c is the train-time method's validation-selected discrimination score but is fully prior-stripped only when $\tau = 1$. The target prior is estimated separately on each natural-validation split and applied only to the corresponding test split. Temperature scaling is fitted on natural-validation NLL after the applicable target-prior correction and then applied unchanged to test logits. Raw, balanced-decision, target-prior-corrected, and temperature-scaled outputs remain separate; a single temperature corrects overall sharpness rather than class-specific prior mismatch.

Balanced Softmax is closely related prior-correction literature: it modifies the softmax denominator using class frequencies and motivates the same distinction between representation learning under a long-tailed source and decision rules for a balanced target (Ren et al., 2020). It is not an additional tested method; it is included to situate logit adjustment among count-aware softmax objectives.

3.3.9 Center-Focused Affinity Loss (CFAL)

Center-focused affinity loss (CFAL) replaces the linear softmax head with learnable class prototypes and an affinity-based training objective (Mahbub et al., 2024). The motivation is that minority histology classes can remain visually similar in embedding space even when a frozen encoder already separates them coarsely; pulling examples toward compact class centers and penalizing confusion with nearby wrong-class prototypes can therefore improve tail-class boundaries without changing the sampler.

The tested patch implementation keeps the shared MLP trunk that maps a frozen Virchow2 embedding $\mathbf{x}$ to a hidden representation $\mathbf{z} = f(\mathbf{x}) \in \mathbb{R}^{512}$, but replaces the final linear map with one learnable prototype $\mathbf{w}_c \in \mathbb{R}^{512}$ per class. Both embeddings and prototypes are L2-normalized before affinity computation. The Gaussian affinity between example i and class c is

$$a_{ic} = \exp\left(-\frac{\|\hat{\mathbf{z}}_i - \hat{\mathbf{w}}_c\|_2^2}{\sigma}\right), \qquad \hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}, \tag{28}$$

where σ controls how quickly affinity decays with prototype distance. At inference, class scores are $\log a_{ic}$ followed by softmax normalization. The affinities $a_{ic} \in (0, 1]$ are geometric similarity scores, not class probabilities: they are trained with a margin objective and never optimized to satisfy likelihood or calibration constraints. Applying softmax to $\log a_{ic}$ therefore yields a convenient ranking score for argmax decisions, but it should not be read as a calibrated estimate of $P(y = c \mid \mathbf{x})$ without additional post-hoc mapping.

Training optimizes a margin term on affinities rather than cross-entropy on logits. For batch index i with label y_i, define the per-example margin loss

$$\ell_i = \sum_{c \neq y_i} \max\big(0,\ \lambda + a_{ic} - a_{i,y_i}\big), \tag{29}$$

which penalizes wrong-class affinities that exceed the true-class affinity by less than margin λ. A diversity regularizer $R(\mathbf{W})$ encourages prototypes to spread apart by penalizing deviations of pairwise squared distances from their mean. The total objective combines class reweighting, a center-focused local weight, and prototype diversity:

$$\mathcal{L}_{\text{CFAL}} = \frac{1}{B} \sum_{i=1}^{B} \frac{1}{E_{y_i}} (1 - a_{i,y_i})^{\gamma}\, \ell_i + R(\mathbf{W}), \tag{30}$$

where E_c is the effective number of training examples in class c derived from class count n_c and hyperparameter β, and γ up-weights examples whose true-class affinity remains low. Unlike the CE plus soft-F1 and soft-MCC hybrids, CFAL does not use balanced oversampling: it is categorized as an algorithm-level patch method because it changes the head and loss while keeping the natural training distribution.

The reference defaults are $\lambda = 0.1$, $\beta = 0.999$, and $\gamma = 2.0$ following Mahbub et al.; all three are held fixed and the affinity bandwidth σ is the tuned control selected on validation; it sets the metric scale of Equation (28) rather than a rebalancing strength, so no value of it recovers ordinary CE. The paper's $\sigma = 130$ was calibrated for un-normalized AlexNet FC features; on L2-normalized 2,560-dimensional Virchow2 vectors that scale is far too large and collapses all pairwise affinities, so the tested implementation evaluates a range matched to the normalized embedding geometry. CFAL is evaluated only in the patch regime because the source method assumes instance-level labels rather than weak slide-level bags.

3.4 Hybrid Sampling–Loss Methods

Hybrid methods combine data-level and algorithm-level changes. The Scholz-style patch objectives below adapt the metric-derived losses of Scholz et al. for imbalanced medical classification (Scholz et al., 2024). Both tested variants use class-balanced oversampling together with metric-derived losses, so their effect cannot be attributed only to a new loss or only to a new sampler.

3.4.1 Scholz-Style CE + Soft F1

The intuition is that CE optimizes the log-probability of the correct class one example at a time, whereas macro F1 evaluates a different object: the balance between precision and recall at class level. Under class imbalance, this distinction matters because a model can reduce CE substantially by fitting frequent classes while still leaving tail-class recall or precision weak.

The obstacle is that ordinary F1 is computed from hard counts after prediction, so it is not directly usable as a gradient-based training loss. The Scholz-style soft-F1 objective replaces those hard counts with probability-weighted counts inside each minibatch. For one class, a confident correct prediction contributes almost one soft true positive, a confident prediction for the wrong class contributes to soft false positives, and low probability on a true class contributes to soft false negatives. Formally, for one-hot labels y_i and predicted probabilities $\hat{y}_i$:

$$\mathrm{TP} = \sum_i y_i \hat{y}_i, \quad \mathrm{FP} = \sum_i (1 - y_i)\hat{y}_i, \quad \mathrm{FN} = \sum_i y_i(1 - \hat{y}_i). \tag{31}$$

These terms keep the familiar meaning of a confusion matrix but make every component differentiable with respect to the model probabilities. They then define soft precision, soft recall, and soft F1:

$$\mathrm{F1}_{\mathrm{soft}} = \frac{2\,\mathrm{precision}_{\mathrm{soft}}\,\mathrm{recall}_{\mathrm{soft}}}{\mathrm{precision}_{\mathrm{soft}} + \mathrm{recall}_{\mathrm{soft}}}. \tag{32}$$

The multiclass implementation applies this construction separately to each class and averages the resulting classwise terms. This macro-style averaging is the imbalance-aware part: each class contributes one term regardless of how many samples it has in the full dataset. The tested objective does not replace CE completely; it adds the metric-derived term to CE so that the objective retains a likelihood term while also pushing the minibatch toward higher macro soft F1:

$$\mathcal{L}_{\mathrm{CE+F1}} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{aux}} \left(1 - \frac{1}{C} \sum_{c=1}^{C} \mathrm{F1}_{\mathrm{soft},c} \right). \tag{33}$$

The auxiliary weight $\lambda_{\mathrm{aux}} > 0$ controls the contribution of the soft-F1 term and is selected on validation. This loss is trained with balanced sampling on the patch manifest. That design is important for interpretation: the loss tries to optimize a macro-style class metric, while the sampler ensures that tail classes occur often enough in minibatches for their soft-F1 terms to provide a regular training signal. The retained CE term should not be read as preserving proper-scoring-rule behaviour. CE alone is minimized by the true conditional probabilities, but the soft-F1 term is not, and their sum is therefore no longer a proper scoring rule: its minimizer trades probability accuracy against the metric at a rate set by λ_{aux}. The same holds for the soft-MCC hybrid below. Both are consequently reported with calibration measured rather than assumed.

3.4.2 Scholz-Style CE + Soft MCC

The soft-MCC variant uses the same principle as soft F1—replace hard evaluation counts with differentiable probability masses—but starts from the Matthews correlation coefficient (MCC). Whereas F1 for one class depends mainly on that class's positive predictions, MCC is built from the full confusion matrix and therefore reacts to false positives, false negatives, and true negatives at the same time. In binary problems it is often preferred in imbalanced settings because it can remain informative when one class dominates the dataset: accuracy can look high while predictions are still systematically biased, but MCC penalizes both kinds of mistakes jointly.

Binary MCC as a correlation coefficient. For one class treated as positive, write TP, TN, FP, and FN for the four confusion-matrix cells. The standard MCC is

$$\mathrm{MCC} = \frac{\mathrm{TP} \cdot \mathrm{TN} - \mathrm{FP} \cdot \mathrm{FN}}{\sqrt{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}}. \tag{34}$$

The numerator compares co-occurrence of correct positive and correct negative decisions with co-occurrence of the two error types; the denominator scales by how many examples fall into each margin of the confusion matrix. MCC equals +1 for perfect predictions, 0 for chance-level association, and can become negative when predictions are worse than random. This makes it a correlation-style score rather than a prevalence-weighted rate.

Multiclass MCC in a minibatch. For C target classes, the multiclass extension treats labels and predictions as C-dimensional vectors per example. Let $Y \in \{0,1\}^{N\times C}$ be the one-hot label matrix and $\widehat{Y} \in [0,1]^{N\times C}$ the corresponding softmax probabilities, with rows $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$. Three aggregate quantities describe how mass is distributed inside the batch:

$$s = \sum_{i=1}^{N}\sum_{c=1}^{C} Y_{ic}\widehat{Y}_{ic}, \qquad t_c = \sum_{i=1}^{N} Y_{ic}, \qquad p_c = \sum_{i=1}^{N} \widehat{Y}_{ic}. \tag{35}$$

Here s is the total probability-weighted agreement between labels and predictions: each example contributes $\hat{y}_{i,y_i}$, the predicted mass on its true class. The terms t_c and p_c are the batch totals of true and predicted mass for class c. They play the role of marginals in the confusion-matrix calculation. If predictions collapse onto a few head classes, the vector $(p_1, \ldots, p_C)$ becomes skewed even when argmax accuracy looks acceptable; MCC is sensitive to that mismatch because it compares the joint alignment s with what would be expected from the marginals alone.

The multiclass MCC used in training can be written as a normalized covariance between the label matrix and the prediction matrix:

$$\mathrm{MCC}_{\mathrm{mc}} = \frac{Ns - \sum_{c=1}^{C} t_c p_c}{\sqrt{N^2 - \sum_{c=1}^{C} p_c^2}\,\sqrt{N^2 - \sum_{c=1}^{C} t_c^2}}. \tag{36}$$

The numerator $Ns - \sum_c t_c p_c$ increases when true and predicted mass line up example by example and decreases when correct predictions are no better than what the marginal totals would suggest by chance. The two square-root terms measure how spread out the batch-level label totals and prediction totals are across classes. They do not, however, make few-class batches harmless. A minibatch whose examples all carry the same label has $t_c = N$ for that class and zero elsewhere, so $N^2 - \sum_c t_c^2 = 0$ and the denominator vanishes; the prediction-side term degenerates in the same way when the softmax outputs collapse onto one class. MCC is undefined in exactly those batches rather than merely large, which is a case the implementation has to handle explicitly. Equation (36) reduces to the familiar binary form in (34) when $C = 2$.

Soft MCC loss. Hard MCC replaces $\widehat{Y}$ by one-hot argmax predictions, which again blocks gradient-based training. The soft-MCC objective keeps the structure of (36) but plugs in softmax probabilities directly:

$$\mathrm{MCC}_{\mathrm{soft}} = \frac{Ns - \sum_{c=1}^{C} t_c p_c}{\sqrt{N^2 - \sum_{c=1}^{C} p_c^2}\,\sqrt{N^2 - \sum_{c=1}^{C} t_c^2}}. \tag{37}$$

Every term in (35) and (37) is differentiable with respect to the logits that produce $\widehat{Y}$. A confident correct prediction increases s; probability placed on the wrong classes increases some p_c while leaving the matching t_c unchanged, which lowers the numerator; and overly peaked prediction vectors reduce the prediction-side denominator term, limiting the score even if a few examples are classified correctly. The degenerate batches identified above are handled numerically rather than left to the algebra: both spread terms are floored at $\epsilon_{\mathrm{spr}} = 1 \times 10^{-8}$ before the square roots are taken, and a further $\epsilon = 1 \times 10^{-8}$ is added to the resulting denominator, so a single-class batch contributes a bounded term with a vanishing gradient instead of a division by zero. Soft F1 is stabilized the same way, with $\epsilon_{\mathrm{F1}} = 1 \times 10^{-16}$ added to the denominator

$2\,\mathrm{TP} + \mathrm{FP} + \mathrm{FN}$ of Equation (32), which is zero for a class absent from the batch and unpredicted in it; that class then contributes a full unit of loss, which is the intended macro-averaging behaviour. Because balanced sampling supplies the minibatches for both hybrids, single-class batches are rare at the batch sizes used, but the floors are part of the specification rather than a fallback. With that in place, the training objective turns a score to be maximized into a loss to be minimized and adds it to CE:

$$\mathcal{L}_{\mathrm{CE+MCC}} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{aux}}\,(1 - \mathrm{MCC}_{\mathrm{soft}})\,. \tag{38}$$

The auxiliary weight $\lambda_{\mathrm{aux}} > 0$ has the same role and validation grid as in the soft-F1 hybrid. This variant is also evaluated only in the patch regime and is trained with class-balanced oversampling. The soft-F1 and soft-MCC variants therefore ask a focused question: once tail classes are regularly presented to the optimizer, does adding a differentiable version of a class-balanced evaluation metric improve the controlled patch classifier beyond CE and simple balanced sampling?

3.4.3 Odd-K-Out (OKO) Set Learning

Standard cross-entropy optimizes each example independently, which has two relevant consequences under class imbalance. First, the gradient signal for a tail class arrives only when that class is sampled, so rare examples can be overwhelmed by frequent ones. Second, hard-label cross-entropy encourages logit values to diverge without bound, which is a known driver of overconfidence and poor calibration (Guo et al., 2017). OKO (Muttenthaler et al., 2024) addresses both problems simultaneously by replacing per-example cross-entropy with a set-level objective. Calibration is inherently a property of a model's output distribution over a collection of examples, and training on sets is a natural way to expose the model to that distributional structure during learning.

Set construction. A training set S of size $k + 2$ is constructed as follows. A pair class y' is sampled uniformly from $[C]$, equalizing class frequency in expectation regardless of the training distribution—analogous to balanced sampling but applied at the set level rather than the sample level. Two examples x'_1, x'_2 are drawn from the class y', and k distinct odd classes $y'_3, \ldots, y'_{k+2}$ are sampled without replacement from $[C] \setminus \{y'\}$. One representative example x'_j is drawn from each odd class y'_j. The resulting set $S = (x'_1, \ldots, x'_{k+2})$ therefore always contains exactly one matched pair and k distractors from different classes.

Set-aggregated loss. For a model f_θ parameterized by θ, OKO defines the aggregated logit vector as the sum of individual per-example outputs over the set members:

$$f_\theta(S) = \sum_{i=1}^{k+2} f_\theta(x'_i). \tag{39}$$

Prediction on the aggregated logits uses ordinary softmax. The pair class y' appears twice in the set and its two logit contributions therefore reinforce each other, raising the aggregate score for y' relative to any single odd class. The aggregation is a sum, not an average: adding members raises the magnitude of $f_\theta(S)$ rather than damping it, so the calibration behaviour cannot be attributed to pooling shrinking the logits. It arises instead from the objective the sum is placed inside. Predicting the pair class from an aggregate that also carries k odd-class contributions means the gradient rewards a member's logits only insofar as they help the set-level decision, and Muttenthaler et al. (2024) derive from this an implicit entropic regularization of the learned representation—a bias toward less extreme, higher-entropy outputs—which is what they link to improved calibration under hard labels.

The main training loss predicts the pair class from the aggregated logits:

$$\ell_{\text{OKO}} = -\log \text{softmax}(f_\theta(S))_{y'}. \tag{40}$$

Following the main-text configuration of the original paper, the implementation also trains an auxiliary classification head g_θ that shares the trunk with f_θ and predicts the first odd class y'_3:

$$\ell_{\text{aux}} = -\log \text{softmax}(g_\theta(S))_{y'_3}. \tag{41}$$

The total per-set loss is $\ell_{\text{OKO}} + \ell_{\text{aux}}$. The auxiliary head is discarded at inference, and single-example evaluation proceeds exactly as for any other classifier: the model accepts one patch feature vector and returns the logits from the main head. There is no inference overhead or architecture change at test time.

OKO belongs in the hybrid sampling–loss category because the set construction couples a balanced class-sampling policy with a modified training objective; neither component is meaningful on its own. Balanced sampling via uniform pair-class selection ensures that all C classes contribute equally to the gradient signal in expectation, which directly addresses the data-level imbalance problem. The aggregated set-level loss then introduces an implicit regularization on logit magnitude. Together these are not separable in the same way that standard cross-entropy and a WeightedRandomSampler are separable. OKO is evaluated in the patch feature regime only; the set-construction mechanism does not extend naturally to the attention-MIL bag encoder of the WSI-bag regime.

The control k is the number of odd classes in a set, and therefore the set cardinality $k+2$; it is the tuned control selected on validation. It fixes how many distinct classes a single aggregate must discriminate among, not a degree of averaging and not a rebalancing strength. The paper's ablation experiments show that accuracy and calibration are insensitive to the specific value of k, and its main configuration uses $k = 1$ together with the auxiliary odd-class head of Equation (41); $k = 1$ is also the computationally cheapest option, though a moderate set size can work best in practice without requiring the largest sets. Because a set consumes $k + 2$ forward passes, the comparison against CE is made at a matched number of optimizer updates, as in the source, rather than at matched examples seen.

3.5 Representation and Architecture-Level Methods

Representation-level methods try to make rare classes more separable in feature space rather than only changing sampling or per-example loss weights. This family is especially relevant for computational pathology, where WSI labels supervise bags of many unlabeled instances and class imbalance can occur both across slides and within a positive slide (Juyal et al., 2024).

3.5.1 Classifier Re-Training (cRT)

Classifier re-training (cRT) separates representation learning from classifier fitting (Kang et al., 2020). Stage one trains the ordinary CE model on the observed imbalanced training distribution. Stage two freezes the learned representation, discards and reinitializes the final linear classifier, and fits only that classifier using class-balanced sampling. If $h_{\widehat{\theta}}(x)$ is the frozen representation learned in stage one, cRT estimates

$$(\widehat{W}, \widehat{\mathbf{b}}) = \arg\min_{W,\mathbf{b}} \mathbb{E}_{(x,y)\sim q_{\text{bal}}} \left[-\log \text{softmax}\big(W h_{\widehat{\theta}}(x) + \mathbf{b}\big)_y \right], \tag{42}$$

where q_{bal} gives classes equal expected exposure. This isolates whether imbalance primarily distorts the decision boundary after a useful representation has already been learned.

In the patch regime, stage one learns the common MLP head on frozen Virchow2 embeddings; cRT freezes its learned hidden representation and retrains a reinitialized linear output layer with balanced patch sampling. In the WSI regime, stage one learns the instance projection and attention aggregator; cRT freezes both components and retrains only a reinitialized bag classifier with balanced slide sampling. Validation selects the stage-two checkpoint, and its compute is reported in addition to stage-one training. The original cRT study considers several classifier normalization variants; the tested implementation uses only the reinitialized linear-classifier variant specified here.

3.5.2 SC-MIL

SC-MIL extends supervised contrastive learning to MIL by applying the contrastive objective to bag representations instead of assigning noisy slide labels to individual patches (Juyal et al., 2024). The motivation is that a slide label usually does not describe every patch in a bag. A cancer-type label can be correct for the slide while many instances contain background, normal tissue, technical noise, or weakly informative morphology. Applying contrastive learning directly to instances would therefore treat many patches as positives or negatives using labels they do not truly carry. SC-MIL avoids that mismatch by contrasting the aggregated slide-level representation.

The attention-MIL backbone first maps bag B_i to an embedding $\mathbf{b}_i = \mathbf{b}(B_i)$. A projection head g then produces normalized contrastive features $\mathbf{r}_i = g(\mathbf{b}_i)$. In the contrastive view, each bag in a minibatch is an anchor. Bags with the same cancer-type label are positives because their slide-level representations should become closer, while bags from other classes act as negatives because their representations should remain separated. For anchor i, bags of the same class form the positive set $P(i)$, and the supervised contrastive term is

$$\mathcal{L}_{\text{SCL}} = -\frac{1}{\sum_i |P(i)|} \sum_i \sum_{p \in P(i)} \log \frac{\exp(\mathbf{r}_i^\top \mathbf{r}_p / \tau)}{\sum_{a \neq i} \exp(\mathbf{r}_i^\top \mathbf{r}_a / \tau)}. \tag{43}$$

The numerator rewards similarity to same-class bags. The denominator compares that similarity against all other bags in the minibatch, so the loss favors representations where class clusters are compact and mutually separated. The temperature τ controls how sharply these similarities are weighted. The source reference is $\tau = 1.0$; it is retained in the validation grid together with lower temperatures.

SC-MIL does not rely on contrastive learning alone. The contrastive term shapes the representation space, while CE trains the final class predictor. The tested variant combines them through a linear curriculum:

$$\mathcal{L}_t = \beta_t \mathcal{L}_{\text{SCL}} + (1 - \beta_t)\mathcal{L}_{\text{CE}}, \qquad \beta_t = 1 - \text{progress}_t. \tag{44}$$

Early training therefore emphasizes class-structured bag embeddings, when the classifier head is still unstable. As training progresses, β_t decreases and more weight moves to CE, so the learned representation is tied back to the actual slide-level classification objective. Class-aware batching draws at least two slides for each represented class whenever class support permits; otherwise an anchor would have $|P(i)| = 0$ and contribute no contrastive term. Training logs the number of valid anchors, ordered positive pairs $\sum_i |P(i)|$, and represented classes per batch. The normalization in Equation (43) uses the actual number of ordered positive pairs, so unequal class multiplicities do not silently change the scale of the contrastive objective.

3.5.3 MDE-Inspired Dual-Expert MIL

The MDE-inspired variant uses distribution-specific experts and a cross-expert consistency term for long-tailed WSI classification (Ling et al., 2025). The full source method combines a dual-expert ensemble with multimodal distillation from a pathology foundation model. The tested implementation includes only the ensemble component: a shared attention aggregator, two expert classification heads, and optional logit consistency under the same frozen Virchow2 bags as the other MIL methods. It omits CONCH-based multimodal distillation and is therefore not called a faithful MDE-MIL reproduction.

The shared trunk matches plain attention MIL: each bag $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ is encoded instance-wise, attention weights are computed with a linear scorer and softmax over instances, and the weighted sum yields a bag embedding $S(B) \in \mathbb{R}^d$. Two separate feed-forward experts then map S to logits:

$$E_U(S) = W_U\, \phi_U(S) + b_U, \qquad E_B(S) = W_B\, \phi_B(S) + b_B, \tag{45}$$

where ϕ_U and ϕ_B are two-layer MLPs with ReLU and dropout. Expert U is trained on the natural slide distribution and expert B on a class-balanced slide distribution.

Each training step draws one minibatch from each distribution. The unbalanced loader shuffles slide bags as in MIL CE; the balanced loader uses the same inverse-frequency weights as balanced-sampling MIL. For batches (B_U, y_U) and (B_B, y_B), the classification loss is

$$\mathcal{L}_{\text{cls}} = \text{CE}\big(E_U(S(B_U)), y_U\big) + \text{CE}\big(E_B(S(B_B)), y_B\big), \tag{46}$$

and the cross-expert consistency term penalizes disagreement on the *same* bag embedding:

$$\mathcal{L}_{\text{con}} = \text{MSE}\big(E_U(S(B_U)), E_B(S(B_U))\big) + \text{MSE}\big(E_B(S(B_B)), E_U(S(B_B))\big). \tag{47}$$

The total objective is $\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{con}}\mathcal{L}_{\text{con}}$. The reference default is $\lambda_{\text{con}} = 0.25$, matching the Camelyon+-LT default reported by Ling et al.; λ_{con} is selected on validation. Setting $\lambda_{\text{con}} = 0$ defines the prespecified *dual experts without consistency* component ablation: both natural and balanced experts remain, but only their classification losses are optimized. Both experts share the aggregator throughout training.

At validation and test time, each slide bag is evaluated once in natural order with mean-logit ensemble inference:

$$\hat{\mathbf{z}}(B) = \tfrac{1}{2}\big(E_U(S(B)) + E_B(S(B))\big). \tag{48}$$

MDE-inspired dual-expert MIL is categorized as a hybrid WSI method because it couples two sampling distributions with an optional consistency regularizer rather than changing only the sampler or only the per-example loss.

4 Summary

This report places the tested roster on two orthogonal axes—the signal a method consults to infer that a class or example needs help (prevalence, support in its nominal and independent readings, difficulty, diversity) and the level at which it intervenes (sampling, objective, prior correction, representation)—and defines every method in the input form it consumes, with CE and MIL CE as the no-mitigation references (Table 2). Most collected methods read prevalence; the remaining signals are covered by class-balanced CE, pilot-difficulty weighted CE, AREA and semantic-scale weighted CE, and by independent-support weighted CE, a thesis construction. Six of them share one intervention level—a mean-one class weight on unmodified CE—which is what makes the signal comparison interpretable.

Where a source method could not be reproduced as published, the deviation is stated with the method and the result labelled accordingly. Output semantics are recorded where an intervention changes them, prior-corrected and temperature-scaled scores being kept separate from raw ones. These definitions are what the evaluation of discrimination, probability quality, and mitigation recovery presupposes.

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